CMZ Phase 2 Automatic analysis report

Digital Image Processing | Wanikkasit Nopthiraitthikun and Thanachai Naksomboon

1. Main results and scope

31 original videos across four cameras contain 109.84 camera-hours. The completed historical run sampled 1,319 two-second camera windows, or 43.97 camera-minutes. Shared nominal times yield 348 fused points. These are sampled-point proposals, not whole-day behavior durations.

Proposal	Points	Share
Walking	53	15.23%
Feeding	9	2.59%
Unknown	286	82.18%
Total	348	100.00%

Active proposals: 62/348 (17.82%). Verified Inactive: unavailable. Unknown contains 198 conditional possible-shelter points. Missing detections do not confirm rest or sleep.

All historical table totals were recomputed and matched. The source records and independent data-integrity check are included.

2. Workflow and trained model

Inventory and preserve source files. Decode by FFmpeg presentation timestamps.

Use nine existing reviewed clips to derive 37 training segments: 30 Feeding and 7 Walking. Keep historical source-offset limitations visible.

Extract 18 motion features from 96 x 96 grayscale full frames using Farneback optical flow and frame differences.

Train a Random Forest with 200 trees, depth 8, balanced class weights, minimum two samples per leaf and seed 42.

Use pretrained MegaDetector V5A to propose animal presence. Apply the trained RF only when detection and classifier gates permit.

Sample one two-second window per five-minute stratum. Combine matching camera times. Export per-video, fused-time and hourly results.

Evaluation boundary

The existing Walking development check recovered 18 of 30 seconds as Walking. The remaining 12 seconds were Unknown. This 60% development ratio is not overall accuracy. Independent multi-class precision, recall and F1 remain unavailable. Software tests and arithmetic checks are separate evidence.

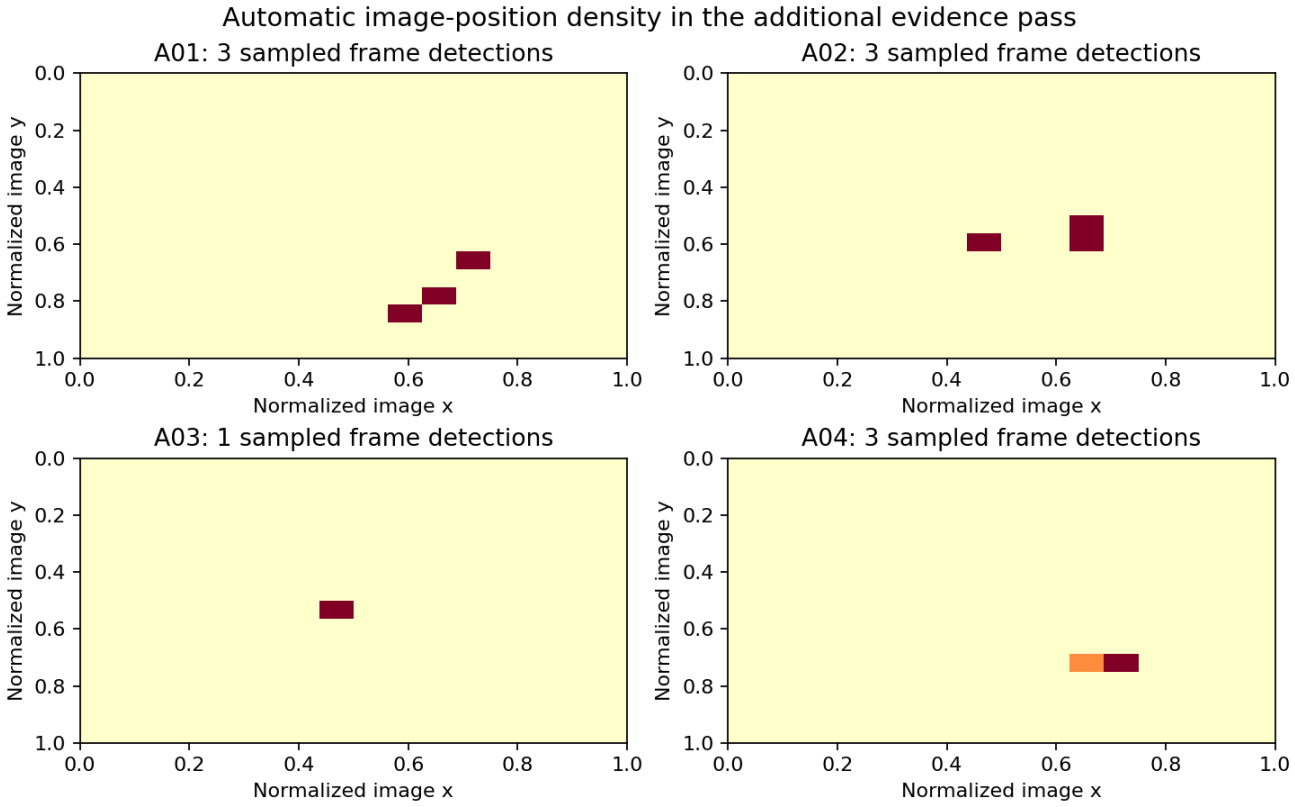
3. Additional automatic evidence

Processed 77 of 77 planned windows with 0 errors. Evaluated 548 detector frames and found animal proposals in 10 frames at threshold 0.4.

Scope: 53 saved convenience audit clips plus 24 eight-second systematic windows from one additional original video. Keep these results separate from the 348 historical sample points. The reconstructed RF uses the encoded reviewed clips and is a distinct model.

Cohort	Automatic behavior proposals
additional_systematic_windows	{'Unknown': 24}
saved_convenience_audit	{'Unknown': 53}

Motion proxies: {'Unknown': 76, 'Moving proposal': 1}. A stationary box is not a verified resting or sleeping animal. Perspective, missed detections and box jitter affect image-plane movement.



Position density counts detected centroids in sampled frames. It does not measure room occupancy or dwell time.

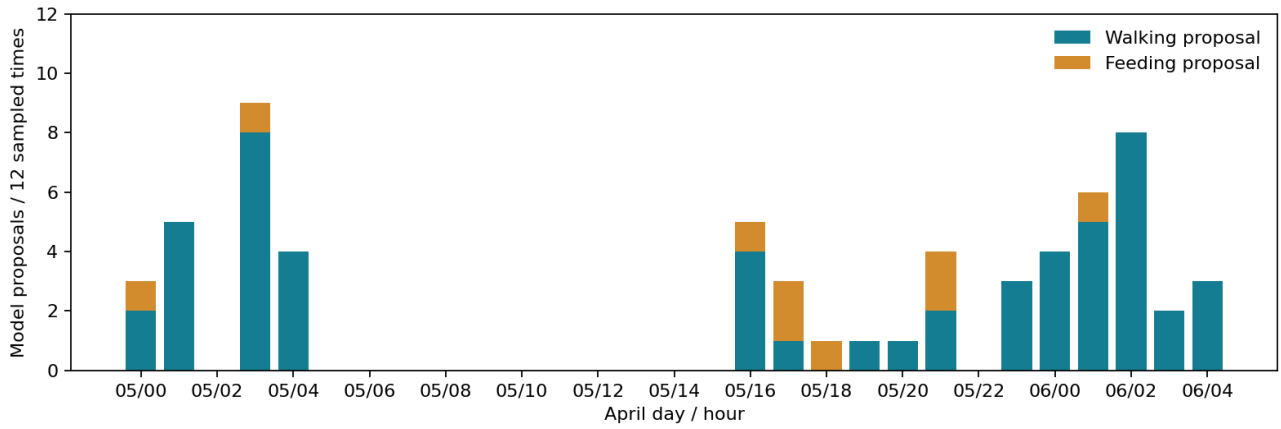
4. Image improvement experiment

Twelve saved audit stills were selected before the paired experiment. Some inputs already contain red ROI outlines. Raw means no additional photometric transform. Counts show proposals, not correct detections or validated accuracy gains.

Variant / size	Images detected at 0.4	At 0.2
raw / 640	1/12	3/12
raw / 1280	9/12	9/12
clahe / 1280	8/12	9/12
gamma_0.7 / 1280	9/12	9/12

CLAHE adjusts luminance contrast (clip limit 2.0, 8 x 8 tiles). Gamma 0.7 brightens the frame. Larger detector input changes object scale. These operations cannot restore lost image detail and can increase noise or false positives. Historical result tables remain unchanged.

Hourly sampled proposals



Each hourly bin has 12 nominal sampled times. Unknown is omitted from these bars. Zero Active proposals does not establish inactivity.

5. Requirement coverage and review links

Requirement	Current evidence
Train selected suitable data	Implemented pilot. 9 reviewed clips; 37 derived segments; two classes only
Explain selected AI / image model	Documented. Pretrained MD detector plus trained RF behavior pilot
Design / evaluate Active-Inactive output	Partially demonstrated. Active proposals available; verified Inactive unavailable
Provide source code / annotation link	Packaged. Source offsets retain explicit unverified provenance

Phase 1 continuity: same problem, setup, behavior definitions and visibility distinction. The implementation changes the detection/classification method. Climb, verified Resting, room-use calibration and continuous full-day durations remain unsupported by these results.

Fast trace during questions

Open OPEN_THIS_FIRST.md in the repository. Each question links directly to the matching table, model settings, annotation register, software test log or data-integrity check. REQUIREMENTS_TRACE.csv maps requirements to slide numbers and evidence.

Code: github.com/Wanikkasit65/civet_project Historical revision: 1163e2091505f35026d71a53c8eb3b7ce454e469 Detector guide: megadetector.readthedocs.io/en/latest/userguide.html

The full Markdown report contains exact settings, cohort definitions and source references. No independent accuracy claim is made, and the old Phase 1 96.3% figure is not reused as a Phase 2 result.